

Paper Under Submission, Abstract

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Weaver: Foundation Model Training over AI-RAN Compute Infrastructure

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ABSTRACT

The emergence of AI-RAN architectures for future 6G networks envisions deep integration between radio access network (RAN) infrastructure and AI workloads. While prior work primarily focuses on AI-for-RAN and AI-on-RAN, the potential of AI-and-RAN, sharing RAN compute infrastructure with non-RAN (AI) workloads, remains largely unexplored. In this paper, we investigate the feasibility of leveraging GPU-accelerated RAN infrastructure to support opportunistic training of foundation models (FMs), one of the most compute-intensive AI workloads. We conduct the first comprehensive characterization of spare GPU capacity in GPU-accelerated RAN systems at both micro-scale (per-slot within a cell site) and macro-scale (across sites). Our analysis reveals substantial underutilized compute resources, approximately 40–85%, that are temporally bursty within sites yet spatially complementary across sites. To efficiently exploit this opportunity in a RAN-safe manner, we present **Weaver**, a system that enables FM training to opportunistically share GPU resources with latency-critical RAN workloads while preserving RAN performance. Weaver adopts a RAN-first design with coordinated elasticity across system layers. It introduces a RAN-centric spare-compute controller embedded in the MAC scheduler to stabilize spare GPU capacity, alongside a two-level elastic FM training framework that adapts to dynamic resources across and within sites. Evaluations on an O-RAN-aligned 5G NR prototype and multi-site testbed show that Weaver reduces spare-compute fluctuations by up to $4.9\times$ and achieves $2.1\text{--}4.2\times$ higher training throughput than baseline approaches while harvesting up to 83% of available spare compute.

MY CONTRIBUTION

- **Characterisation and profiling.** Ran the GPU profiling and roofline analysis behind the spare-capacity characterisation, showing that RAN kernels are memory-bandwidth-bound while FM training is compute-bound, the asymmetry that makes SM-level co-location viable.
- **Controller evaluation.** Evaluated the RAN-centric spare-compute controller, including instrumenting the MAC scheduler to measure per-deadline completion and confirm that live RAN performance is preserved while capacity is harvested.
- **Large-scale experiments.** Carried out the large-scale distributed FM training runs on the multi-site testbed used in the evaluation.